

Appendix: full model comparison of Autoencoders

Model	TPR @ FPR=1e-5 [%]				TPR @ FPR=1e-5 [Events]				AUC [%]			
	$A \rightarrow 4\ell$	$h^\pm \rightarrow \tau\nu$	$h^o \rightarrow \tau\tau$	$LQ \rightarrow b\tau$	$A \rightarrow 4\ell$	$h^\pm \rightarrow \tau\nu$	$h^o \rightarrow \tau\tau$	$LQ \rightarrow b\tau$	$A \rightarrow 4\ell$	$h^\pm \rightarrow \tau\nu$	$h^o \rightarrow \tau\tau$	$LQ \rightarrow b\tau$
dense_ae_1	0.032	0.112	0.035	0.006	18	849	244	19	83.46	89.29	74.27	82.01
dense_ae_2	0.486	0.048	0.029	0.004	272	362	201	12	87.88	88.36	73.52	84.56
dense_ae_3	0.132	0.075	0.030	0.002	74	568	208	8	88.45	88.53	73.50	85.29
dense_ae_4	0.118	0.076	0.026	0.004	66	579	178	14	84.44	89.24	74.64	83.19
dense_ae_5	0.139	0.148	0.045	0.012	78	1125	313	40	87.68	88.95	74.06	83.48
dense_ae_6	0.100	0.186	0.081	0.011	56	1415	560	39	88.84	89.33	75.42	86.62
dense_ae_7	0.070	0.126	0.039	0.004	39	960	273	13	85.04	89.59	74.11	82.53
dense_ae_8	0.332	0.085	0.034	0.004	186	647	238	13	87.62	88.07	72.49	84.63
dense_ae_9	0.204	0.009	0.011	0.004	114	72	77	13	81.77	83.52	69.32	82.36
dense_ae_10	3.582	0.086	0.077	0.049	2005	657	531	167	90.06	87.47	72.45	82.99
dense_ae_11	1.093	0.115	0.043	0.012	612	875	297	42	87.87	88.39	74.85	85.30
dense_ae_12	0.177	0.066	0.027	0.008	99	505	188	28	90.11	87.55	70.90	81.71
dense_ae_10_loss0511	1.333	0.063	0.043	0.018	746	476	295	60	89.95	85.74	69.66	81.51
dense_ae_10_loss111	1.092	0.080	0.048	0.015	611	606	333	51	89.38	87.14	71.47	83.45
dense_ae_10_loss20505	3.606	0.062	0.057	0.032	2018	470	393	109	88.00	85.80	67.03	78.65
dense_ae_10_mse	0.663	0.029	0.038	0.007	371	220	260	25	88.18	84.22	71.07	82.48
dense_vae_0	0.490	0.069	0.035	0.018	274	524	239	60	89.19	90.28	74.95	85.88
dense_vae_1	0.189	0.083	0.030	0.008	106	630	206	27	87.81	90.31	74.97	85.97
dense_vae_2	0.390	0.096	0.031	0.003	218	730	213	11	79.58	88.13	68.01	74.51
dense_vae_3	0.127	0.095	0.027	0.004	71	725	186	14	81.57	87.08	68.69	76.52
conv2d_ae_0	0.197	0.076	0.019	0.005	110	581	128	16	88.18	87.95	74.02	85.28
conv2d_ae_1	0.470	0.046	0.056	0.011	263	353	387	37	90.49	89.05	73.34	82.81
conv2d_ae_2	0.005	0.010	0.005	0.000	3	74	35	0	53.22	62.52	53.53	50.37
conv2d_ae_3	0.432	0.033	0.044	0.011	242	251	305	39	91.84	89.51	75.22	84.98
conv2d_ae_4	0.007	0.039	0.015	0.001	4	299	103	4	78.80	86.80	74.26	81.90
conv2d_ae_4_mse	0.104	0.093	0.022	0.004	58	710	151	15	87.03	83.10	69.19	80.29
conv1d_ae_0	0.306	0.088	0.062	0.026	171	666	430	88	89.89	89.03	71.89	83.76
conv1d_vae_0	0.331	0.125	0.080	0.039	185	954	555	134	90.08	88.38	73.38	85.27